Journal of Material Cycles and Waste Management
https://doi.org/10.1007/s10163-025-02457-4

REVIEW

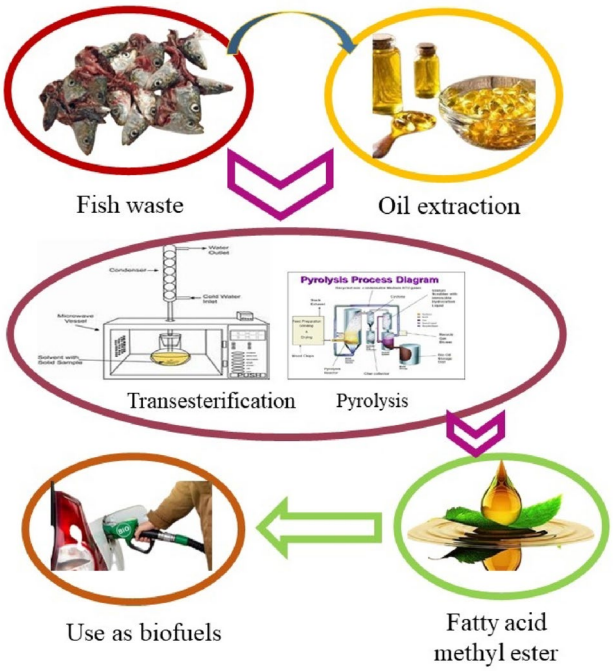
Transforming fish waste into fuel: A sustainable bioenergy solution

Esha Razzaq[1] · Syed Makhdoom Hussain[1] · Shafaqat Ali[2] · Ameer Fawad Zahoor[3] · Hamda Azmat[4] · Adan Naeem[1] · Ebru Yilmaz[5] · Eman Naeem[1] · Muhammad Amjad[1]

Received: 5 June 2025 / Accepted: 3 December 2025
© The Author(s), under exclusive licence to Springer Nature Japan KK, part of Springer Nature 2025

Abstract
The global fishing industry generates millions of tons of waste annually, posing environmental and health risks. The valorization of fish waste can potentially lower the expenses related to saving disposal and generate valuable products by extracting valuable compounds like collagen, protein, gelatin, chitin, and oil. The increasing global demand for sustainable energy has led to exploring alternative bioenergy sources. Recent research has revealed that fish discards are viable alternatives for producing biofuels with economic and environmental benefits. This comprehensive review describes the latest innovations in the production of biofuel from fish waste and also highlights their efficiency, challenges, and environmental impact. Additionally, the economic feasibility and sustainability of fish-waste-derived biofuels are discussed in comparison to conventional fossil fuels and other biomass sources. The key benefits of biofuel production from fish waste include being a renewable energy source, reducing waste, lowering greenhouse gas emissions, and being cost-effective. This review underscores the need for further research, technological advancements, and policy support to optimize fish waste utilization for sustainable bioenergy solutions.

Graphical abstract



Keywords Fish waste · Oil extraction · Biofuels · Fossil fuels · Transesterification · Environmental friendly

Extended author information available on the last page of the article

Published online: 22 December 2025

Introduction

The world's growing population has increased demand for protein-rich meals, making fish vital to global food security [1]. The recovery of valuable components from fish, such as proteins, oils, and bioactive compounds, is a critical strategy for resource valorization. However, even after getting valuable components, significant amounts of residual fish waste remain unsuitable for direct reuse due to compositional, economic, or logistical limitations [2]. Energy recovery from this residual waste is therefore necessary not only to prevent environmental pollution but also to contribute to the production of renewable energy. Conversion of fish waste into fuels such as biodiesel, hydrotreated vegetable oil (HVO), or bioLPG maximizes overall waste utilization, supports the growing global energy demand, and aligns with the principles of a circular bioeconomy, complementing traditional material recycling strategies [3].

According to the FAO [4], global aquatic animal production, including finfish and shellfish, reached 177.8 million metric tons, with finfish accounting for approximately 87–90% of the total. Production reached a record high of 177.8 million metric tons worth USD 406 billion in 2020 [4]. This review is limited to finfish waste because of the significant variation in waste composition among aquatic organisms fish waste is rich in proteins and lipids, whereas shellfish and seaweed wastes primarily consist of chitin, minerals, or polysaccharides because biofuel production requires high lipid or fatty acid content, which shellfish and seaweed generally lack [5]. All mentioned fish waste in this review refers to by-products obtained from finfish processing, encompassing heads, viscera, frames, skin, scales, and bones [6]. Fish is high in protein, vitamin D, phosphorus, and calcium. Fish has 15–30% protein, 0–25% fat, and 50–80% moisture. Fish composition is mostly determined by species and age [7]. De-shelling, degutting, fin and scale removal, and filleting produce solid waste because 70% of fish and seafood are treated before sale [8]. Unintentionally caught fish is another major economic issue that produces a huge amount of trash. Throwing away trash is an enormous waste of resources that affects the long-term use of marine biological resources and marine environments. Oxygen depletion arising from the consumption of aerobic bacteria and decomposing organic waste, combined with fish discarding, has an immense impact on the coastal ecology throughout the food chain [9]. Fish industry waste disposal methods, primarily landfilling and incineration, cause substantial harm to the environment, human health, and the economy [10].

About 20% of fish waste is processed and used, but the rest is dumped, burnt, or discharged into the marine environment as trash, causing health, environmental, and financial issues [11]. Thus, fish waste processing requires novel methods to protect the environment and fish sustainability. Several programs and laws have been designed to address this issue. The European Common Fishery Policy (CFP) and the UK Environmental Protection Act are used worldwide to control and mitigate fish waste [12].

Although, despite its high quality, only a portion of the fish is consumed, resulting in a growing accumulation of waste that harms the ecosystem. Disposing of these wastes in an open location without treatment causes them to have an undesirable odor. Previous discussions indicated that this may exacerbate environmental and water contamination as well as other municipal problems [13]. Quick and creative responses are needed to address the growing problem of fish waste. Responsible management of fish by-products that lessen their impact on the environment while simultaneously producing goods with added value is crucial for fostering economic development and the circular economy [14]. Reusing and recycling fish byproducts is the goal of fish waste management, which includes processes such as treatment, specification, control, prevention, and supervision. Goals for sustainable consumption and production were set by the UN in 2015 as part of the Sustainable Development Agenda. These include protecting marine and maritime environments and drastically reducing global food waste per capita [15]. To encourage sustainability in the aquaculture and fishing industries, it is crucial to make use of waste products and discards. A circular economy strategy that makes use of waste products can be implemented in the fish sector, and this is becoming more apparent. To further prevent the depletion of natural resources, this strategy might be extended to include fishery by-catch, which would aid in the search for ecologically viable solutions to worldwide economic and environmental problems. Sustainable development and environment friendly growth in the maritime and oceanic industries are the goals of the Blue Growth program run by the European Commission [16].

The process of fish waste valorization could reduce costs because expenses incurred in collecting, transporting, and safely managing waste to prevent environmental pollution-while simultaneously creating value by extracting valuable components. Enzymes, peptides, oligosaccharides, oils, collagen, gelatin, vitamins, minerals, and more can be recovered cheaply and effectively from fish waste as byproducts. Byproducts of fish contain oil and fat content that varies from 0.3% to 45% (w/w) and is influenced by biological and environmental factors such as species, age, sex, nutrition, health, and seasonality [17]. Revitalized products from fish waste can reduce waste. Numerous studies are isolating bioactive compounds from fish waste using simple or complex technologies to use as animal feed, food components, and packaging [18]. High-value items including proteinaceous

meals utilized in medicine formulation, sports drinks, and food supplements, are made from fish waste [19].

As we know world is facing rising energy demands, depleting fossil fuel resources, and environmental degradation, securing energy supplies has emerged as a critical need. Coal, oil, gas, and diesel, collectively known as fossil fuels, are primary energy sources for various industries, such as transportation, business, manufacturing, and agriculture. Fossil fuels supply around 90% of energy needs today [20]. The growing energy needs and expenses emphasize shifting from traditional fossil fuels such as petroleum and diesel to more sustainable options. Because after burning, these fuels release harmful substances into the atmosphere, leading to health problems and air pollution. The main releasing pollutants are sulfur dioxide, CO_2, and nitrogen oxides [21]. Every day, more and more gasoline and diesel are being used, and the direct link between fuel-burning emissions and raising price, emphasizing how important it is to take action to reduce them [22].

These issues can be solved effectively with alternative fuels that provide a comprehensive solution, integrating cost-effectiveness, sustainability, renewability, and minimum environmental impact [23]. Rising oil prices and climate change concerns have sparked a global wave of interest in biofuel production from biomass resources, driving academic innovation and research in renewable energy solutions. Because biofuels provide a sustainable and environment friendly energy alternative to conventional fuels. They are characterized by their non-toxic, biodegradable, cost-effective, and abundant nature. Biofuels require careful consideration of production factors, including catalyst type, temperature, reaction duration, and biomass chemical makeup [24].

Biofuel represents a key component in the transition towards a sustainable energy future, particularly in transportation, where it can substantially reduce CO_2 emissions [25]. Fish waste is transferred into valuable resources through eco-friendly and profitable means, including biodiesel, biogas, and other biofuels production [17]. Biofuels, produced from various feedstocks like waste oils, agro-waste, and animal fats, provide a cleaner alternative to fossil fuels. Biofuels boast environmental benefits, including biodegradability, non-toxicity, and significantly lower emissions of greenhouse gases, carbon monoxide, nitrogen oxides, sulfur oxides, unburned hydrocarbons, and particulate matter [26]. Despite advances in converting fish waste into valuable products and biofuels, challenges remain in scalability, cost-effectiveness, regulatory support, and byproduct utilization. This review aims to analyze current strategies for fish waste valorization, highlight technological opportunities, control rising prices, reduce environmental pollution, and provide insights for future research and industrial adoption.

Bioactive molecules from fish processing

Fish bodies vary by species, nutritional state, and health, with protein ranging from 15 to 30%, fat from 0 to 25%, moisture from 50 to 80%, and ash. Bones contain high amounts of minerals like calcium, phosphorus, etc [27]. The main by-products from fish processing and fisheries by-catch are rich sources of valuable bioactive compounds like proteins, fats, enzymes, collagen, and gelatin [28].

Oil

The global fish oil market is predicted to grow 5.79% from $1,905.77 million in 2019 to $2,844.12 million in 2027. Every year, the European Union generates around 120,000 tons of fish oil. Fish oil contains fat-soluble vitamins and polyunsaturated fats. The parts of fish, including the skin, liver, head, frames, fins, meat, and the spaces around the belly, often contain 2–30% fat and are ideal for making fish oil. Fish species, by-products used for fat extraction, age, sex, health state, harvest season, and extraction process are the primary factors that influence fish oil composition [28,29]. Oil is extracted from fish by-products through techniques like microwave-assisted extraction, Soxhlet method, etc. Fish oil capsules are used as dietary supplements. The incorporation of fish oil as an enrichment material for long-chain ω−3 fatty acids in foods such as refined soybean salad oil, apple beetroot juice, flavored yogurt, and orange drinks [30]. Byproducts of fish processing emerged as a significant source of fish oil. Byproducts of fish are used at the industrial level because they contain a high amount of oil with omega-3 fatty acids for biofuel production [31]. Oil from fish waste is a renewable resource of environmentally sustainable biodiesel. The suitability of fish waste oil for biodiesel production has been extensively studied and confirmed [32].

Proteins

Fish waste contains a lot of readily digested, nutritious proteins due to the availability of essential amino acids [33]. The protein content in fish skeletons rich in lysine, valine, and phenylalanine, ensures high nutritional value and digestibility [34]. The bioactive peptides found in fish protein are involved in metabolic processes, including the ability to hold water, absorb oil, gel, foam, and emulsification [27]. Fish protein has a balanced amino acid content compared to other animal protein sources. Enzymatic hydrolysis converts fishery waste into nutritious proteins and amino acids, replacing disposal with utilization rather than being discarded as waste. Adding fish protein into less healthy foods is a great way to boost their nutritional value [35].

Enzymes

Fishery by-products, particularly viscera, offer significant enzyme potential due to high activity and stability across various pH levels. These enzymes provide significant industrial potential, particularly in seafood processing for effective collagen extraction [36]. Pepsin is the predominant protease in fish viscera, playing a vital role in protein digestion. Pepsin, trypsin, chymotrypsin, and collagenase enzymes comprise a large fraction of bioactive chemicals in fishery by-catch and by-products, with commercial extraction potential. Trypsin and trypsin-like enzymes successfully extract from various fish species. Chitinases and chitosanases are found in the digestive system and other organs. These enzymes aid in recovering chitin and chitosan from marine waste for various biological uses. Employing fishing by-products to extract proteolytic enzymes promotes industrial innovation and environmentally sustainable waste management [37].

Collagen

Collagen is one of the most abundant animal proteins in skin, bones, tendons, cartilage, and connective tissues. Type I collagen, the primary component of marine collagen, is from 28 types and has a triple helix structure [32]. Fish skin, bones, fins, scales, jellyfish, starfish, and sea cucumber connective tissue are used to obtain marine collagen, and 70–80% of dry matter of skin contains collagen. Fish scales make up 4% of the yearly 18–30 million tons of fish waste, and are another promising and low-cost source of marine collagen. Collagen, fat, lecithin, scleroprotein, and hydroxyapatite are found in fish scales [38]. It is mostly used in environmental protection, biofuel production, medicine, and food applications. Collagen obtained from marine animals has a similar or slightly lower molecular weight than other mammals [39].

Using chemicals to remove collagen from fish scales takes time. Chemical collagen extraction is described in many papers [14]. Scientists are interested in collagen extraction methods for fish scales. Collagen has high biodegradability biocompatibility, low antigenicity, and cell growth potential properties. Collagen has texturization, thickening, and gel-forming properties [40].

Gelatin

Fish gelatin is a biopolymer consisting of protein, mineral salt, and water. Collagen and gelatin are two forms of the same protein. The polypeptide chains in gelatin are arranged in a repeating (Gly-X-Y) pattern, where Gly is glycine, X is commonly proline, and Y is hydroxyproline. Changes in gelatin's physical and chemical properties can occur as a result of acylation, esterification, deamination, cross-linking, and interactions with acids and bases, among other chemical treatments. Hydrolysis of collagen produces gelatin which is a tasteless, colorless, solid material. Gelatin is a denatured, partially hydrolyzed form of collagen. It originates from the thermal denaturation of collagen. When collagen changes into gelatin by hydrolysis the molecular forces between collagen break. Gelatin is characterized as bovine, porcine, or fish gelatin but fish gelatin is mostly used due to its applications and functional qualities like antioxidant, antihypertensive, antimicrobial, wound healing promotion, and also have film-forming qualities which abundantly used in food sector areas for packaging and medicines [28].

Chitin and Chitosan

Pure chitin is a white or off-white amino polysaccharide made of N-acetyl-β-D-glucosamine units and monomers linked via β (1, 4) links. It is odorless and tasteless. Its structural similarities with cellulose lead it to classify as the second most prevalent natural biopolymer. The phylum Arthropods including crustaceans and insects are the main source of chitin because their exoskeleton is made up of it, fish scales and other crustaceans like mollusks, shrimp, etc. and fish waste are also good sources of chitin. Common carp scales are the best source of chitin [41]. Demineralization, deproteination, and decolorization of waste are the main steps for obtaining chitin. Chitooligomers, chitinase, chitosanase, antioxidants, and antidiabetic chemicals are obtained from chitin and used for cancer treatment [42].

By product of chitin is chitosan linear polysaccharide chitosan is formed by β (1, 4) glycosidic linkages that link deacetylated and acetylated d-glucosamine units. Alkali solutions deacetylate chitin to make chitosan for commercial use. The main advantage of chitosan is its solubilization property because of the amino group in its structure [43]. Chitosan, a common marine polysaccharide, is receiving attention in food packaging because it is used as a natural food preservative due to its antibacterial properties, it effectively suppresses bacteria, yeasts, and fungus and also promising factor in food packaging due to its non-toxic qualities, biodegradability, and food preservation performance [44].

Sources of biofuels

A wide range of feedstocks, including vegetable oils, soap stock, food waste, fungi, algae, and animal fats, have been explored for biofuel production (Fig. 1). However, among these, fish processing waste stands out as a sustainable and non-edible resource, rich in lipids and proteins, that avoids

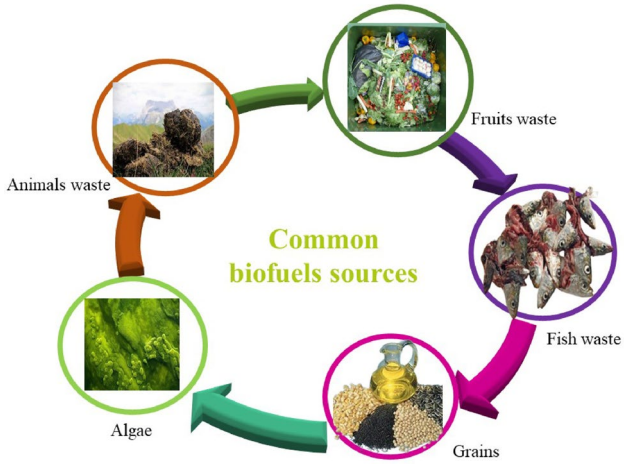


Fig. 1 Different sources used for biofuel production [3]

competition with food crops and reduces environmental burdens [45]. Generated in large quantities from heads, viscera, frames, and skin, fish waste offers significant potential for biodiesel and biogas production, while simultaneously lowering disposal costs for the seafood industry. Compared with conventional feed stocks like vegetable oils, animal fats have poor fuel properties, algae and fungi remain costly to cultivate, and food waste is inconsistent. In contrast, fish processing waste is abundant, non-edible, and lipid-rich making it a sustainable, low-cost feedstock for biofuel production while reducing environmental burdens. It provides the dual benefit of renewable energy generation and waste management, although variability in composition and seasonal availability remain challenge for large-scale application [46].

Fish oil extraction methods

In recent years, a limited researchers have explored the generation and efficacy of biodiesel derived from fish oil sourced from fish waste. The Marine Institute of Memorial University of Newfoundland, Canada, has reported the production of biodiesel from salmon oil [47]. But before the production of biofuels, one main thing is to extract oil from fish waste through different methods. The three main components of fish raw materials are solids, oil, and water. Researchers constructed several techniques and devices to enhance fish oil extraction, increasing yields and quality. Numerous experimental publications, reviews, and patents have fully demonstrated the method's techniques and effective industrial-scale use [48]. Fish oil is extracted through different techniques.

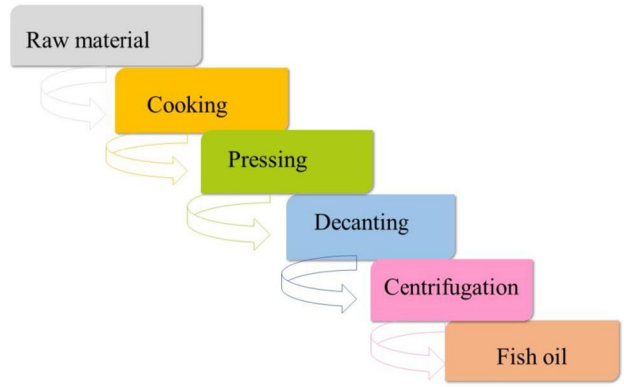


Fig. 2 Wet reduction processing steps [49]

Wet reduction

Wet reduction is the preferred approach for industrial-scale fish oil extraction since it is safer and more environment friendly. This method aims to efficiently separate and isolate the solid, oil, and water components through unit operations [2]. The process involves cooking whole fish or fish waste with steam or hot water, followed by pressing or centrifugation to isolate oil from solids (Fig. 2). According to Taati et al. [49], a specific protocol involves defrosting frozen fish by-products for 12 h, then mixing 200 g of by-products with 1 L of water and heating at 95 °C for 15 min. After cooking, the mixture is pressed, and the resulting liquid is separated into oil and water via centrifugation. This method yields both fish oil and protein-rich meal, although it requires significant energy and careful handling.

Soxhlet method

This technique used for extracting lipids and oils from various substances is the soxhlet apparatus with a liquid solvent. Ethyl acetate, hexane, and petroleum ether are examples of liquid solvents often used for lipid extraction. The soxhlet extraction process is used to extract oil from animal sources, including fish. The soxhlet extraction method involves a continuous solid-liquid extraction, which involves repeatedly adding solid materials to a liquid mixture for extraction [50]. For this, a distillation flask is used to reflux the solvent, which is then filtered or washed through the lipid-containing substance multiple times. The soxhlet method offers several key benefits: enhanced extraction through repeated sample-solvent interaction, improved throughput via parallel processing, and reduced processing steps, as filtration is unnecessary due to heat transfer. This technique is better suited for laboratory-scale or small-scale extractions, and its commercial or industrial applications are limited [51].

Microwave-assisted extraction (MAE)

Microwave-assisted extraction (MAE) uses non-ionizing electromagnetic radiations with a frequency range of 300 MHz to 300 GHz [52]. Electromagnetic radiation creates an electric field, making particles move and molecules rotate. This movement produces heat and evaporates water molecules from the processing material. As water evaporates, pressure builds up inside cells, which causes cell walls to break apart and allows bioactive compounds to escape. This approach has been developed to extract bioactive compounds and other constituents from by-products for example oil from fish by-products [53].

Microwave-assisted extraction is an automated extraction method that is environmentally friendly and uses a low amount of energy. The main benefits of the Microwave-assisted extraction process are that it reduces extraction time, requires less organic solvent (water), and is more reproducible, it depends on the capacity of the extraction solvent to absorb microwave radiation and the interaction between the sample and the solution [54]. Microwave frequency, timing, and sample particle size affect its efficiency, and the study reveals equivalent or better efficiency in fish oil extraction despite limited use [55].

Supercritical fluids (SFs)

For many common processes, such as the extraction of metalic ions and essential oils, supercritical fluids are used as solvents. Because of its mild critical temperature and pressure, supercritical ($scCO_2$) is the most employed supercritical fluid [56]. Supercritical fluids have a dual nature, characterized by liquid-like density and gas-like properties such as low viscosity, high diffusivity, and compressibility, at particular temperature and pressure circumstances. By modifying pressure and temperature, the density-dependent polarity of supercritical fluids can be set up, facilitating the selective extraction and concentration of diverse chemicals. Due to its non-toxic, non-corrosive, and non-flammable properties, $scCO_2$ is a safe option for extraction, and its gaseous form allows for easy elimination without residual contaminants. The low-temperature environment and oxygen displacement by $scCO_2$ ensure the stability of thermos table substances and prevent omega-3 fatty acids from oxidation. By modifying the temperature and pressure, this method selectively extracts the desired component from the supercritical fluid [57].

Enzymatic hydrolysis

Enzymatic hydrolysis for oil extraction is a promising method; more research is present on enzymatic hydrolysis than other methods. In enzymatic hydrolysis, the enzymes come from different sources (Fig. 3). Enzymatic hydrolysis in fish processing promotes protein extraction, producing essential nutritional supplements. Adding enzymes from outside the body makes the digestion process easier to control and perform. So, it is the best way to get oil and protein fish industrial waste [58]. Enzymes and the fish used in the process must be safe for food. If the enzymes come from microbes, they cannot serve as pathogens. Mostly alkaline proteases are employed for hydrolysis due to their superior performance to acidic proteases. Before extraction, the exogenous enzymes must be turned inactive by fixing the pH. Oil extraction is affected by protease activity, concentration, pH, temperature, and particle size [55].

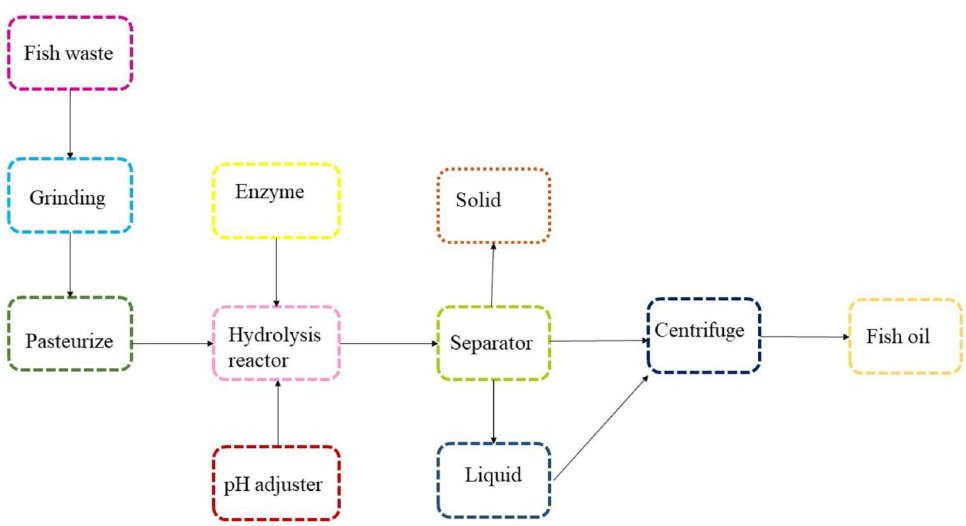


Fig. 3 Steps of typical hydrolysis [55]

Table 1 Merits and demerits of fish oil extraction techniques [59]

Techniques	Merits	Demerits	References
Wet reduction	- Use less solvent - Lowers biomass drying energy requirements	- High temperatures and pressures may lead to oil oxidation - Extracted lipids may be of lower quality than dried material	[59]
Soxhlet method	- Simple and easy to perform - Quick process	- Time-consuming - Requires extensive solvent usage	[22]
Microwave-assisted extraction (MAE)	- Reduces solvent usage and extraction time - Operates at lower temperatures	- Inefficient with volatile solvents - High energy consumption	[60]
Supercritical fluid extraction (SCFE)	- Operates at low temperatures - Provides high extraction yields	- Requires sophisticated equipment and high temperatures	[61]
Enzymatic hydrolysis	- It does not require organic solvents	- High enzyme costs and limitations in process control	[62]

Table 2 Comparative analysis of fish oil extraction methods

Method	Fish Source	Production%	Time	References
Wet reduction	*Pangasius* waste	14–15%	2–3 h	[63]
Soxhlet extraction	*Pangasius* waste	16%	6–8 h	[63]
Microwave-assisted extraction	Mixed fish waste	12–18%	19 mints	[64]
Supercritical CO_2	Yellowtail viscera	18–23%	1–2 h	[65]
Enzymatic hydrolysis (Alcalase)	Mackerel heads or whole fish waste	75%	4 h	[58]
Enzymatic hydrolysis (Protease)	Atlantic salmon by-products	13–19%	No fixed time	[66]

Merits and demerits of fish oil extraction techniques

Each fish oil extraction technique has its merits and demerits. The choice of method depends on the specific application, fish type, and desired oil quality. Understanding these factors is crucial for sustainable and efficient fish oil production (Table 1).

Comparative analysis of fish oil extraction methods

Table 2 highlights the comparative analysis of fish oil extraction methods, focusing on production percentage and time, based on various studies.

Enzymatic hydrolysis and $scCO_2$ extraction are the most effective techniques for fish oil recovery, offering high yields and superior oil quality, though both involve higher costs and longer processing. Microwave- and ultrasound-assisted methods offer faster, eco-friendly alternatives, albeit with moderate yields, whereas Soxhlet extraction remains effective but less sustainable due to its solvent use.

Biofuels

Biofuels (biodiesel, biogas) represent key sustainable and renewable energy options, replacing traditional fossil fuels [67]. Biofuels fall into two categories: primary and secondary. Primary Biofuels are derived from natural, organic materials, main sources are wood, wood chips, pellets and organic waste and are used in household cooking, industrial heating and electricity generation. Secondary biofuels are derived from cultivated biomass and generate liquid fuels for transportation and industrial use [68].

Biodiesel production technologies

The term biodiesel describes mono-alkyl esters produced from oil and fat. Biodiesels are synthesized chemically and serve as a replacement for fossil fuels. Production methodologies, influential factors, and fish oil attributes collectively impact biodiesel quality [69]. Following technologies are pivotal in converting oil into biodiesel [70]. However, these methods, such as the use of micro-emulsions with short-chain alcohols, Pyrolysis, Esterification of free fatty acids with alcohols, using solids acids catalysts or bio-enzymes, and Transesterification of triglycerides catalyzed by bases, acids, or enzymes, are mostly used for this purpose.

Pyrolysis

Pyrolysis produces biochar, bio-oil, and non-condensable gases by heating decomposing biomass without oxygen. Each product's yield depends on pyrolysis parameters such as temperature, heating rate, residence time, reactor setup, and catalysts [71]. Modifying these parameters can change the product distribution during pyrolysis, typically at atmospheric or slightly higher pressure [72]. The process usually occurs in an inert gas environment, however the vacuum or hydro-pyrolysis approach provides a stressed hydrogen

atmosphere. Primary and secondary reactions occur during this process. In primary pyrolysis, heat-disrupt chemical bonds in raw materials. Secondary pyrolysis includes cracking, dehydration, condensation, etc. Another factor affecting pyrolysis is feedstock particle size, smaller particles increase liquid product during pyrolysis [73]. Despite its benefits, pyrolysis faces challenges, making it less preferred for extraction. Oxygen and water in pyrolyzed oil decrease its energy density. Blending fish oil with pyrolyzed oil improves its energy value. Pyrolysis provides an environmentally friendly, waste-free oil conversion process. Further refinement is necessary for this method [74].

Transesterification

The transesterification process is one of the most common ways to make biodiesel. It involves reacting fats/oils with alcohol and producing fatty acid alkyl esters (FAAE) and glycerol. This technique substitutes the glyceride group with hydroxyls (alcohol), yielding glycerol as a desirable byproduct. The process reduces the viscosity of bio-oils without affecting their cetane number or energy content. Various catalysts like acidic alkaline or many others are on an industrial scale [75]. In addition to being a commonly recognized chemical process for producing biodiesel, this reaction is often carried out at high temperatures and is very easy to implement compared to other ways [76].

Transesterification in the presence of acid catalysts, including sulfuric and sulfonic acids. This method is ideal for organic substrates. Sulfuric and sulfonic acids are the primary choices, while hydrochloric and phosphoric acids are homogeneous acid catalysts used in transesterification. Heterogeneous acid catalysts sulfuric ion exchange resin and sulfonic mesostructure are the solid acid catalysts used for the transesterification reaction [77]. Solid catalysts are preferred over liquid catalysts in acid-catalyzed transesterification due to their numerous benefits such as rapid transesterification, lowest product contamination due to catalyst separation after reaction, heterogeneous catalyst biodiesel not requiring washing, and can be recycled. Higher alcohol-to-oil ratios, optimized reaction temperature, and catalyst concentration are needed to produce biodiesel with acidic catalysts. Biodiesel production through this method requires high alcohol-to-oil molar ratios and elevated temperatures [78].

Rosson et al. [79] tried zinc oxide-ionic liquid catalysts to make biodiesel from fish oils, their optimal 65 °C and 7-hour biodiesel output was 92%. Slow reaction rates made acid catalysts less effective for biodiesel generation from fish discards. Its slow pace and temperature requirements have restricted the conversion of fish waste to biodiesel on an industrial scale. This method is unsuitable for producing biodiesel from fish waste [80].

The most effective catalysts for transesterification are base catalysts, such as NaOH and KOH, which enable the reaction to occur at lower temperatures, pressures, and reaction times. There are two types of alkaline catalysts like acid catalysts, homogeneous and heterogeneous, each with its own set of efficiency and performance. NaOH and KOH are effective for making biodiesel from fish oil [81]. Kudre et al. [82] studied the effect of homogeneous alkaline catalysts on fish biodiesel yield from catfish waste oil and found that yield was achieved by utilizing a potassium hydroxide catalyst for (1 h at 32 °C) and production of 96% fish biodiesel utilizing KOH catalyst with 1:0.5 fish oil to alcohol ratio at 55 °C temperature and 1 h reaction time. Homogeneous alkaline catalysts have issues with saponification and glycerol extraction from fish biodiesel. Heterogeneous base catalysts solve that. Calcium zinc oxide (CaZnO), calcium oxide-nickel oxide, and a few others are used as heterogeneous alkaline catalysts but they have high water and moisture absorption rates, making this method ineffective [83]. Fish bones and scales are also used as heterogeneous alkaline catalysts, which are cost-effective and do not need to depend on commercial catalysts [84]. Alkali-based catalysts are a cost-efficient option, with lower pricing. The capacity to recycle alkali-based catalysts enhances the sustainability of biodiesel production. By reducing biodiesel costs, this advancement promotes environmental sustainability. The method is a promising solution for future biofuel production. Because of cheap alkaline catalysis, it is one of the most commonly employed processes on an industrial scale. More work is necessary to examine commercial-scale production, utilizing the advantages of low-temperature requirements, alcohol solubility, and rapid processing [85].

Parameters optimize biodiesel production

Alcohol-to-oil proportion

The quantity of fish oil and alcohol utilized in biodiesel production is a crucial component that affects the resultant products. Some studies have revealed that the alcohol-to-oil ratio changed throughout the transesterification process to examine how this changes the reaction rate and the final quality of the biodiesel made from fish. The normal ratio for the transesterification reaction is 3:1. The amount of alcohol used in fish oil is frequently monitored because it affects the total price. Optimal alcohol-to-oil ratios speed up reactions [86]. Transesterification can make alkyl esters of fish oil fatty acids from methanol, butanol, etc. This process favors methanol for its low cost than others. Ethanol from agricultural waste is the next preferred choice. The interface

between oil and alcohol requires adequate dissolution for this process. Transesterification slows when alcohol and oil phases fail to mix [87, 88].

Effective catalyst concentration

Optimizing the catalyst quantity is also crucial. Most transesterification catalysts are acidic, basic, or enzymatic. The number of catalysts greatly affects fish biodiesel yield. Small catalyst amounts required more reaction time to obtain a maximum conversion, but high concentration gave results quickly [89]. Both homogeneous and heterogeneous basic catalysts affected Rohu fish oil biodiesel synthesis, according to Kudre et al. [82] Homogeneous catalysts were sodium and potassium hydroxide, while heterogeneous catalysts were calcium and magnesium oxide. Biodiesel conversion was continuously monitored at 0.5%–1.5% catalyst concentration. When homogeneous catalysts increased above 1.5% by weight, saponification occurred, while heterogeneous catalysts did not. The yield of biodiesel production correlates directly with catalyst levels. Biodiesel yield increases with rising catalyst concentrations [90].

Process temperature and time

Reaction time and temperature are crucial in the transesterification process, as they directly affect the quality, yield, and cost. Biodiesel synthesis requires optimizing reaction time and temperature. Sometimes, reaction molecule collisions accelerate at the correct temperature, favoring transesterification and its products. Temperature plays a key role in this process. Enhancing molecular collisions speeds up the process. Temperature breaks chemical bonds and promotes efficient product generation [91]. Jaiswal et al. [75] found that biodiesel yield reduced after 2 h of Rohu fish oil conversion. During the long reaction period, the esters produced may have been hydrolyzed. Enzyme-catalyzed transesterification cannot use high temperatures because lipases degrade. Acid, base, or lipase catalysts and transesterification processes affect reaction time. Reports show that acid and enzyme catalysts take longer to react. In several tests, methanol evaporates at higher temperatures, affecting fish biodiesel performance.

Comparison of fish oil quality with other feedstock

Fish waste-derived oils have been increasingly recognized as viable biodiesel feedstocks due to their favorable energy content and combustion characteristics. Compared with conventional vegetable oils, fish oils exhibit distinctive fatty acid profiles, being rich in long-chain polyunsaturated fatty acids (PUFAs) such as eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA), alongside oleic and palmitic acids [63]. This composition imparts good cold-flow properties but compromises oxidative stability, making the oil more susceptible to rancidity during storage unless antioxidants or process modifications are applied [92]. Algal oils are also rich in PUFAs, particularly EPA and DHA, and can be a sustainable alternative for biodiesel, although they are generally more expensive to produce and require controlled cultivation [93]. Other animal fats, such as chicken, beef, or pork tallow, are typically high in saturated fatty acids, providing good oxidative stability but poor low-temperature flow properties [94]. In contrast, fish oil biodiesel (≈ 39–41 MJ/kg) is comparable to soybean and palm oil biodiesel (40–41 and 39–40 MJ/kg, respectively), indicating similar energy release during combustion [95]. because vegetable oils such as palm oil, which are high in saturated fatty acids, provide greater oxidative stability but exhibit poor low-temperature flow properties [96]. Non-edible oils like jatropha and castor also demonstrate high cetane numbers and stability but face limitations due to agronomic challenges and processing costs. Recent studies highlight that fish oil biodiesel generally meets international fuel standards when properly transesterified, while offering the added advantage of valorizing processing waste and avoiding food fuel competition [64]. Therefore, although stabilization measures are necessary, fish waste oil represents a sustainable and technically suitable alternative to conventional feedstocks for biodiesel production.

By-products

The biofuel production process yields glycerin and crude glycerol as valuable by-products through multiple chemical pathways. Impure glycerol, tainted by ash, alcohol, fatty acids and trace contaminants are additional products. Impurities impact the physical quality of crude glycerol. Refined glycerol is in highest demand for cosmetics, pharmaceuticals, and food processing. Effective glycerol purification relies on chemical processing, distillation, ion exchange and adsorption [97]. Glycerol, however, has very few uses in the current situation. Therefore, effort should be used to avoid producing too much glycerol. As a result, additional innovative uses of glycerol to produce goods with greater added value have not yet been investigated.

Application of biofuel

Biogas can be produced from fish processing waste in addition to biodiesel. While research has been invested in producing biodiesel, significantly less has been done to produce biogas. Oil-free fish waste (cake) is suitable for biogas production. In addition to oil-free fish waste, manure,

and sewage sludge may also be utilized. The waste products of anaerobic digestion are used as fertilizer Anaerobic digestion offers effective treatment for solid waste disposal. Biogas from anaerobic digestion of solid waste provides eco-friendly energy, reducing greenhouse gas emissions and fossil fuel reliance [98]. Fish waste biogas consists of 50 to 75% methane and 25 to 50% carbon dioxide. Trace quantities of other gases, including nitrogen, oxygen, hydrogen sulfide, and ammonia, are present [21]. Limited research exists on biogas production from fish discards, emphasizing the need for more research on composite fish discards and their composition.

Using heavy waste, anaerobic digestion of biomass is a widely utilized method of generating biogas. Fish wastes can provide high-value organic carbon for methane synthesis, but include a lot of ammonia and nitrogen. Co-digestion allows anaerobic treatment, Co-digestion mixes fish wastes with other wastes for biogas production. Microorganisms break down biodegradable substances through anaerobic processes. Acetogenic and Methanogenic microbes transform organic acids into methane. Anaerobic digestion outputs include nutrient-dense digestate and biogas, which contain methane, carbon dioxide, and minor components such as hydrogen sulfide. The decomposition process is divided into four key steps (Fig. 4) [21].

Advanced biofuel opportunities from fish waste

Beyond biodiesel, fish waste represents a promising feedstock for advanced biofuels such as hydrotreated vegetable oil (HVO) and bioLPG. HVO is produced via hydrotreatment of triglycerides and free fatty acids from fish oil under high-pressure hydrogen with a suitable catalyst [99]. This process saturates double bonds, removes oxygen, and produces a paraffinic, sulfur-free fuel fully compatible with existing diesel engines and distribution infrastructure. Fish oil, due to its high lipid content and low ash, is an efficient feedstock for HVO, providing improved oxidative stability and higher cetane number compared to conventional FAME biodiesel [95].

Similarly, fish waste can be converted into biological liquefied petroleum Gas (bioLPG) through thermochemical processes. This typically involves gasification, where biomass is heated in a low-oxygen environment to produce syngas ($CO + H_2$), followed by catalytic upgrading or methanation to produce propane-rich bioLPG. This gaseous fuel can substitute fossil LPG in domestic, transport, and industrial applications, offering higher energy density and cleaner combustion [100]. These advanced biofuels provide higher energy content, better cold-flow properties, and lower emissions than traditional biodiesel, while valorizing a major fish processing by-product. Developing scalable HVO and bioLPG production from fish waste could enhance the economic and environmental sustainability of the fisheries sector and contribute to renewable energy targets [101].

Comparison between fossil fuel and biofuel

Biofuels have considerable benefits compared to traditional fuels and alternative renewable energy sources, offering a cleaner and more sustainable option [102] (Table 3). Traditional fossil fuels have been replaced by biofuels, which are made from organic sources. Fossil fuels like coal, fuel oil, and natural gas are obtained from non-renewable natural

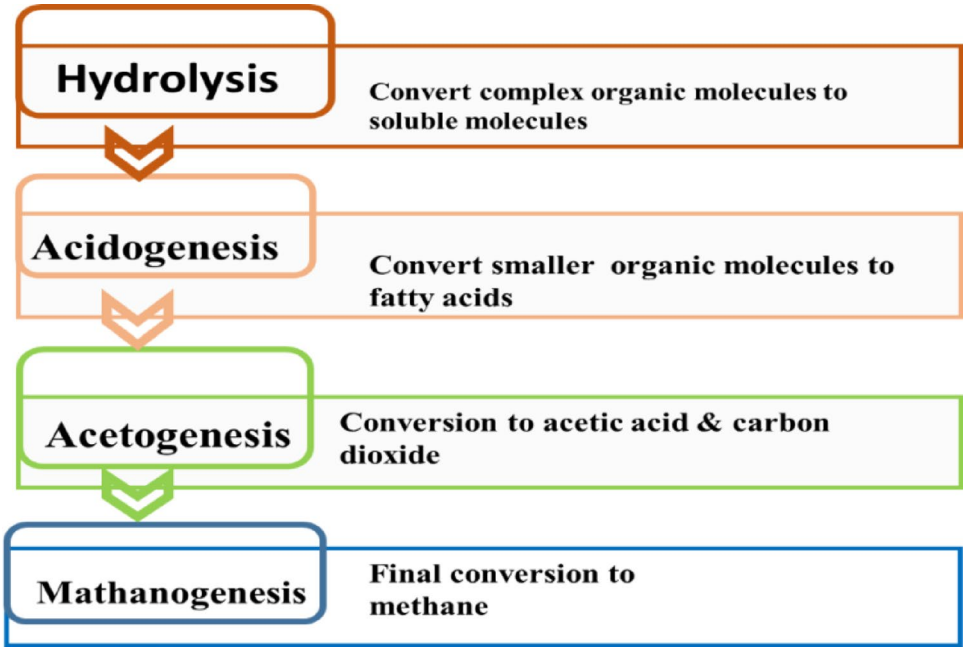


Fig. 4 Steps of aerobic digestion for biogas production [21]

Table 3 Fossil fuels vs. Biofuels [105]

Parameters	Fossil Fuels	Biofuels
Greenhouse gas (GHG) Emissions	High	Lower (varies by generation)
Environmental Impact	Land and water degradation, pollution	Reduced impact; utilizes waste biomass
Renewability	Non-renewable	Renewable, sustainable
Land Use	Exploits natural habitats	Uses marginal land and agricultural waste
Future Potential	Finite, unsustainable	Can potentially eliminate GHG emissions

sources like dead plants and animals [103]. Revenue in the biofuel business is higher than in fossil fuels, which highlights the importance of optimizing production processes, developing better blending capabilities, and increasing global efficiency. According to IRENA, global liquid biofuel consumption is expected 129 billion liters in 2016 to 652 billion liters by 2050, with the transport sector requiring 180 billion liters of biodiesel by 2050 [104].

Scalability and economic feasibility

Fish waste-derived biofuel has strong potential for large-scale production due to the availability of substantial quantities of by-products from the fish waste. Scaling up, however, requires careful consideration of processing methods, energy requirements, labor, and equipment costs. Integrating fish waste into existing biofuel production systems can be both technically feasible and economically viable, offering reduced raw material costs and lower environmental impact compared to conventional feedstocks. Studies have shown that biodiesel produced from fish waste has physicochemical properties similar to conventional diesel, allowing its use in diesel engines with little or no modification [106, 107]. Ali et al. [108] reported that fish waste-derived biodiesel is a sustainable alternative to fossil fuels. Using fish waste reduces environmental pollution, supports the circular economy, and generates renewable energy. Reviewing techno-economic evaluations demonstrated that the conversion of fish waste into biodiesel can be economically viable. For instance, Kara et al. [109] say that it costs US$0.69 per liter to make fuel, while Yahyaee et al. [110] say that it costs US$0.25 per gallon to get the raw materials, which is around US$0.066 per liter. A large amounts of fish waste produced, especially in countries like Pakistan; this approach can enhance energy security and lower reliance on imported diesel. Allocating 50% of domestic fish production to biodiesel would result in 421,000 tonnes of fish waste, providing an estimated 84,000 tonnes of fish oil, which help in decreasing rising prices of diesel and makes a stable economy [111]. Addressing these factors is crucial to ensure that fish waste biofuel can transition from laboratory-scale research to industrial application effectively. Additionally, policy actions make things more doable to reduce greenhouse gas emissions as little as possible, biomass-based fuel has to align with the US Renewable Fuel Standard, which also offers producers valuable credits. Milbrandt et al. [112]. The European Union Renewable Energy Directive gives advanced biofuels made from waste, such as fish-processing waste, tax breaks and other benefits [113]. These economic and policy factors work together to lower the risk of production and make fish-waste biodiesel a more viable long-term alternative to traditional fuels.

Major challenges in biofuels production

Food waste (FW) extends beyond food loss, representing significant shortcomings in land, water, energy, and resource utilization. Ineffective management practices such as incineration, landfill disposal, composting, and using FW as animal feed have aggravated environmental, economic, and societal challenges [114]. Efficient valorization of FW into biofuels through integrated biorefinery systems offers a promising, sustainable alternative. However, several obstacles persist such as techno-economic Assessments like limited studies on the financial and technical feasibility of converting FW to energy hinder commercial adoption, the variability in FW, influenced by regional, temporal, and cultural factors, and necessitates detailed chemical characterization to optimize its use as a biorefinery feedstock. Logistics of Collection and transport of FW is often intermixed with municipal solid waste, and the absence of organized collection systems poses a significant barrier to efficient utilization [115]. Proposed solutions emphasize raising public awareness, establishing structured FW collection systems, and fostering collaboration among food industries, local authorities, and communities. Proximity-based biorefineries, robust sorting mechanisms, and cost-efficient valorization technologies are critical for improving FW-to-biofuel conversion. By implementing these measures, FW management can be transformed into an effective strategy for sustainable energy production [116].

Conclusion

Natural energy sources are decreasing at a rapid rate and natural fuels cause high pollution in the environment. Fish discards offer a new approach to address challenges in biodiesel production, serving as an economical and viable feedstock option. The most biodiesel production from fish oil in the shortest period is done by base-catalyzed transesterification. Potentially useful alternatives to traditional

biodiesel manufacturing methods include lipase extracted from fish viscera and heterogeneous catalysts sourced from fish scales and bones, which show great promise if they can be reused. Using a complete approach to utilize fish discards for biodiesel production would provide reciprocal advantages for the fishing and biodiesel sectors, together with fish markets. This unique technique reduces waste disposal by transforming refuse into a viable, self-sustaining alternative to conventional fuel. By handling both pollution and energy shortages, this strategy provides a long-term solution. Extending the use of biodiesel beyond fish waste oil and into other cost-effective feedstock, such as used cooking oil and animal fats, should be a primary focus of the ongoing study.

In the future, the fish waste biofuel sector will focus on sustainable practices, reducing ecological impact and advancing circular economy approaches. Innovative methods include enzymatic hydrolysis, ultrasonic-assisted extraction, and supercritical fluids extraction to enhance conversion efficiency and automation. Extensive production facilities will develop, rendering fish waste make biofuels more competitive with fossil fuels. Continuous research on optimum conversion pathways, microbial strains, and genetic engineering to improve biofuel production. In future, we have to collaborate with the fisheries, aquaculture, and waste treatment industries to optimize waste utilization and generate revenue.

Acknowledgements The authors gratefully acknowledge the contributions and assistance of the entire team at the Fish Nutrition Laboratory, GCUF, Pakistan.

Author contributions Writing-original draft: Esha Razzaq. Conceptualization, Supervision: Syed Makhdoom Hussain. Review & editing, Conceptualization: Shafaqat Ali, Ameer Fawad Zahoor, Hamda Azmat. Review & editing: Adan Naeem, Ebru Yilmaz, Eman Naeem, Muhammad Amjad.

Funding The authors did not receive support from any organization for the submitted work.

Data availability Not applicable.

Declarations

Competing interest We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, and there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the content of this paper.

Ethical approval Not applicable.

References

1. Cojocaru AL, Liu Y, Smith MD, Akpalu W, Chávez C, Dey MM, Tran N (2022) The seafood system: aquatic foods, food security, and the global South. Rev Environ Econ Policy 16(2):306–326
2. Alfio VG, Manzo C, Micillo R (2021) From fish waste to value: an overview of the sustainable recovery of omega-3 for food supplements. Molecules 26:1002–1025. https://doi.org/10.3390/molecules26041002
3. Vasileiadis V, Zafeiriou E, Karmalis I, Karayannis V, Tsanaktsidis C (2025) Valorization of biodiesel from waste vegetable and animal resources: advancing circular bioenergy through sustainable automotive and marine fuel blends. Int J Energy Sect Manag. https://doi.org/10.1108/IJESM-02-2025-0001
4. FAO (2022) In Brief to The State of World Fisheries and Aquaculture 2022. Towards Blue Transformation. FAO, Rome. https://doi.org/10.4060/cc0463en
5. Kubai RM (2024) Effect of Feed Supplementation with Seaweed and Use of Novel Membrane Bioreactor Wastewater Treatment Method on the Quality of Farmed Nile Tilapia (Doctoral dissertation, JKUAT-CoANRE)
6. Venugopal V (2021) Valorization of seafood processing discards: bioconversion and bio-refinery approaches. Front Sustain Food Syst 5:611835. https://doi.org/10.3389/fsufs.2021.611835
7. Athanasopoulou E, Michailidi A, Ladakis D, Kalliampakou KI, Flemetakis E, Koutinas A, Tsironi T (2023) Extraction of fish protein concentrates from discards and combined application with gelatin for the development of biodegradable food packaging. Sustainability 15(15):12062. https://doi.org/10.3390/su151512062
8. Mishra PK, Gautam RK, Kumar V, Kakatkar AS, Chatterjee S (2021) Synthesis of biodegradable films using gamma irradiation from fish waste. Waste Biomass Valoriza 12:2247–2257. https://doi.org/10.1007/s12649-020-01143-w
9. Lionetto F, Esposito Corcione C (2021) Recent applications of biopolymers derived from fish industry waste in food packaging. Polymers 13(14):2337. https://doi.org/10.3390/polym13142337
10. Araújo CS, Rodrigues AMC, Joele MP (2018) Optimizing process parameters to obtain a bioplastic using proteins from fish byproducts through the response surface methodology. Food Packag Shelf Life 16:23–30. https://doi.org/10.1016/j.fpsl.2018.01.009
11. Araujo J, Sica P, Costa C, Márquez MC (2021) Enzymatic hydrolysis of fish waste as an alternative to produce high value-added products. Waste Biomass Valoriza 12:847–855. https://doi.org/10.1016/j.fpsl.2018.01.009
12. Seixas MJ, Martins E, Reis RL, Silva TH (2020) Extraction and characterization of collagen from elasmobranch byproducts for potential biomaterial use. Mar Drugs 18(12):617. https://doi.org/10.3390/md18120617
13. Yadav A (2024) Fisheries waste a source of valuable resources for sustainable development. Food Process Preserv Technol 2:122–137
14. Coppola D, Lauritano C, Palma Esposito F (2021) Fish waste: from problem to valuable resource. Mar Drugs 19:116. https://doi.org/10.3390/md19020116
15. Nazar A, Hussain SM, Ali S, Zahoor AF, Ghafoor A, Alshehri MA, Rashid E (2025) Sustainable bioplastics production from fish waste: their extraction and applications. Polym Adv Technol 36(2):e70095. https://doi.org/10.1002/pat.70095
16. Lucarini M, Zuorro A, Di Lena G, Lavecchia R, Durazzo A, Benedetti B, Lombardi-Boccia G (2020) Sustainable management of secondary raw materials from the marine food-chain: a case-study perspective. Sustainability 12(21):8997. https://doi.org/10.3390/su12218997

17. Karkal SS, Kudre TG (2020) Valorization of fish discards for the sustainable production of renewable fuels. J Clean Prod 275:122985–123003. https://doi.org/10.1016/j.jclepro.2020.122985
18. Radziemska M, Vaverková MD, Adamcová D, Brtnický M, Mazur Z (2019) Valorization of fish waste compost as a fertilizer for agricultural use. Waste Biomass Valoriz 10:2537–2545. https://doi.org/10.1007/s12649-018-0288-8
19. Shahidi F, Varatharajan V, Peng H, Senadheera R (2019) Utilization of marine by-products for the recovery of value-added products. J Food Bioactives 6. https://doi.org/10.31665/JFB.2019.6184
20. Palaniappan K (2017) An overview of applications of nanotechnology in biofuel production. World Appl Sci J 35:1305–1311. https://doi.org/10.5829/idosi.wasj.2017.1305.1311
21. Yuvaraj D, Bharathiraja B, Rithika J (2019) Production of biofuels from fish wastes: an overview. Biofuels 10:301–307. https://doi.org/10.1080/17597269.2016.1231951
22. Tayib YM, Al–Sheikh F, Shakor ZM (2024) Biodiesel production from fish oil: a review. Biofuels 15:115–128. https://doi.org/10.1080/17597269.2023.2214417
23. Ingle AP, Paralikar P, da Silva SS (2018) Nanotechnology-based developments in biofuel production: current trends and applications. Enzym Resour Renew Energy 289–305. https://doi.org/10.1007/978-3-319-95480-6_11.
24. Mecha NJMF, Dolorosa RG (2024) Distribution and characteristics of spiny Lobster (Panulirus spp) puerulus collection sites in Palawan, the Philippines. J Sci 153(3): 1061-1071. https://www.researchgate.net/profile/Nino-Jess-Mar-Mecha/publication/384118860_Distribution_and_Characteristics_of_Spiny_Lobster_Panulirus_spp_Puerulus_Collection_Sites_in_Palawan_the_Philippines/links/67541cc0ef2dc67228b0719c/Distribution-and-Characteristics-of-Spiny-Lobster-Panulirus-spp-Puerulus-Collection-Sites-in-Palawan-the-Philippines.pdf
25. Demirbas A (2009) Biofuels securing the planet's future energy needs. Energy Convers Manage 50:2239–2249. https://doi.org/10.1016/j.enconman.2009.05.010
26. Correa DF, Beyer HL, Fargione J (2019) Towards the implementation of sustainable biofuel production systems. Renew Sustain Energy Rev 107:250–263. https://doi.org/10.1016/j.rser.2019.03.005
27. Ghaly AE, Ramakrishnan VV, Brooks MS (2013) Fish processing wastes as a potential source of proteins. Amino acids and oils: a critical review. J Microbio Biochem Technol 5:107–129. https://doi.org/10.4172/1948-5948.1000110
28. Kim SK, Mendis E (2006) Bioactive compounds from marine processing byproducts–a review. Food Res Int 39:383–393. https://doi.org/10.1016/j.foodres.2005.10.010
29. Roussos E, Triantaphyllidis G, Ilia V (2024) Status of fishery discards and by-products in Greece and potential valorization scenarios towards a national exploitation master plan. Mar Drugs 22:264. https://doi.org/10.3390/md22060264
30. Joseph TC, Remya S, Renuka V, Jha AK (2019) Fishery industry waste: A resource to be valorised. Veraval Research Centre of ICAR-CIFT, Veraval, Gujarat
31. Jamshidi A, Cao H, Xiao J, Simal-Gandara J (2020) Advantages of techniques to fortify food products with the benefits of fish oil. Food Res Int 137–154. https://doi.org/10.1016/j.foodres.2020.109353
32. Jackson A, Newton RW (2016) Project to model the use of fisheries by-products in the production of marine ingredients with special reference to omega-3 fatty acids EPA and DHA. A report by IFFO and the University of Stirling 12
33. Caruso G, Floris R, Serangeli C, Di Paola L (2020) Fishery wastes as a yet undiscovered treasure from the sea: biomolecules sources, extraction methods and valorization. Mar Drugs 18(12):622. https://doi.org/10.3390/md18120622
34. Milovanovic I, Hayes M (2018) Marine gelatine from the rest Raw materials. Appl Sci 8:2407–2427. https://doi.org/10.3389/fbioe.2021.644595
35. Wangkheirakpam MR, Mahanand SS, Majumdar RK, Sharma S, Hidangmayum DD, Netam S (2019) Fish waste utilization with reference to fish protein hydrolisate-a review. Fish Technol 56(3):169–178
36. Zhou L, Budge SM, Ghaly AE (2021) Extraction, purification and characterization of fish chymotrypsin: a review. Am J Biochem Biotechnol 7:104–123
37. Klomklao S, Kuepethkaew S, Benjakul S, Zhang Y, Simpson BK (2024) Enzymes from fish processing waste materials and their commercial applications. In Fish Waste to Valuable Products.Singapore: Springer Nature Singapore 147–194.https://doi.org/10.1007/978-981-99-8593-7_8
38. Nur RM (2021) The utilitation of fish scale waste as a chitosan. Agrikan J Agribisnis Perikanan 13(2):269–273
39. Sanz B, Albillos Sanchez A, Tangey B, Gilmore K, Yue Z, Liu X, Wallace G (2020) Light cross-linkable marine collagen for coaxial printing of a 3D model of neuromuscular junction formation. Biomedicines 9(1):16–35. https://doi.org/10.3390/biomedicines9010016
40. Salvatore L, Gallo N, Natali ML, Campa L, Lunetti P, Madaghiele M, Sannino A (2020) Marine collagen and its derivatives: versatile and sustainable bio-resources for healthcare. Materials Science and Engineering: C 113:110963–110981. https://doi.org/10.1016/j.msec.2020.110963
41. Irastorza A, Zarandona I, Andonegi M, Guerrero P, de la Caba K (2021) The versatility of collagen and chitosan: from food to biomedical applications. Food Hydrocoll 116:106633–1106644. https://doi.org/10.1016/j.foodhyd.2021.106633
42. Kou SG, Peters LM, Mucalo MR (2021) Chitosan: a review of sources and preparation methods. Int J Biol Macromol 169:85–94. https://doi.org/10.1016/j.foodres.2005.10.010
43. Chen C, Li D, Yano H, Abe K (2019) Insect cuticle-mimetic hydrogels with high mechanical properties achieved via the combination of Chitin nanofiber and gelatin. J Agric Food Chem 67(19):5571–5578. https://doi.org/10.1021/acs.jafc.9b00984
44. Bai C, Wei Q, Ren X (2017) Selective extraction of collagen peptides with high purity from Cod skins by deep eutectic solvents. ACS Sustain Chem Eng 5(8):7220–7227. https://doi.org/10.1021/acssuschemeng.7b01439
45. Pan S, Zabed HM, Wei Y (2022) Technoeconomic and environmental perspectives of biofuel production from sugarcane bagasse: current status, challenges and future outlook. Ind Crops Prod 188:115684–115701. https://doi.org/10.1016/j.indcrop.2022.115684
46. Parida S, Pali HS, Chaturvedi A, Sharma A, Balasubramanian D, Ramegouda R, Varuvel EG (2024) Production of biodiesel from waste fish fat through ultrasound-assisted transesterification using petro-diesel as cosolvent and optimization of process parameters using response surface methodology. Environ Sci Pollut Res Int 31(17):25524–25537
47. Dave D, Ramakrishnan VV, Trenholm S (2014) Marine oils as potential feedstock for biodiesel production: physicochemical characterization. J Bioproc Biotechn 4:1–13. https://doi.org/10.4172/2155-9821.1000168
48. Naseem S, Imam A, Rayadurga AS, Ray A, Suman SK (2024) Trends in fisheries waste utilization: a valuable resource of nutrients and valorized products for the food industry. Crit Rev Food Sci Nutr 64(25):9240–9260. https://doi.org/10.1080/10408398.2023.2211167

49. Taati MM, Shabanpour B, Ojagh M (2018) Investigation on fish oil extraction by enzyme extraction and wet reduction methods and quality analysis. Aquacul Aquari Conserv Legisla 11:83–90
50. Hamzah DF, Sutriana A, Anwar SH (2024) A short review: fish oil extraction methods obtaining oils with high amounts of unsaturated fatty acids. In IOP Conference Series: Earth and Environ Sci 13569(1): 012023).https://doi.org/10.1088/1755−1315/1356/1/012023
51. El-Rahman A, Mahmoud N, Badawy AEK (2018) Extraction of fish oil from fish viscera. Egypt J Chem 61:225–235. https://doi.org/10.21608/ejchem.2018.2798.1230
52. Zhi WJ, Wang LF, Hu XJ (2017) Recent advances in the effects of microwave radiation on brains. Mil Med Res 4:1–14. https://doi.org/10.1186/s40779-017-0139-0
53. Alvi T, Asif Z, Khan MKI (2022) Clean label extraction of bioactive compounds from food waste through microwave-assisted extraction technique-a review. Food Bioscience 46:101580. https://doi.org/10.1016/j.fbio.2022.101580
54. de la Fuente B, Pinela J, Mandim F, Heleno SA, Ferreira IC, Barba FJ, Barros L (2022) Nutritional and bioactive oils from salmon (*Salmo salar*) side streams obtained by Soxhlet and optimized microwave-assisted extraction. Food Chem 386:132778
55. Ivanovs K, Blumberga D (2017) Extraction of fish oil using green extraction methods: a short review. Energy Procedia 128:477–483. https://doi.org/10.1016/j.egypro.2017.09.033
56. Karmakar B, Halder G (2019) Progress and future of biodiesel synthesis: advancements in oil extraction and conversion technologies. Energy Convers Manage 182:307–339. https://doi.org/10.1016/j.enconman.2018.12.066
57. Melgosa R, Sanz MT, Beltránn S (2021) Supercritical CO2 processing of omega-3 polyunsaturated fatty acids–towards a biorefinery for fish waste valorization. J Supercrit Fluids 169:105121–105139
58. Ramakrishnan VV, Ghaly AE, Brooks MS (2013) Extraction of proteins from mackerel fish processing waste using alcalase enzyme. Bioproc Biotechnol 3:2–10. https://doi.org/10.4172/2155-9821.1000130
59. Mofijur M, Rasul MG, Hassan NMS (2019) Recent development in the production of third-generation biodiesel from microalgae. Energy Procedia 156:53–58. https://doi.org/10.1016/j.egypro.2018.11.088
60. Zheng S, Zhang G, Wang H, Long Z, Wei T, Li Q (2021) Progress in ultrasound-assisted extraction of the value-added products from microorganisms. World J Microbiol Biotechnol 37:1–14. https://doi.org/10.1007/s11274-021-03037-y
61. Gedi MA, Bakar J, Mariod AA (2015) Optimization of supercritical carbon dioxide (CO2) extraction of sardine (*Sardinella lemuru Bleeker*) oil using response surface methodology (RSM). Grasas Y Aceites 66(2):e074–e085. http://dx.doi.org/10.3989/gya.0824142
62. Thomas SP, Dikundwar AG, Sarkar S, Pavan MS, Pal R, Hathwar VR, Row TNG (2022) Simultaneous recovery of lipids and proteins by enzymatic hydrolysis of fish industry waste using different commercial proteases. Appl Biochem Biotechnol 164:115–124. https://doi.org/10.1007/s12010-010-9119-5
63. Rahman N, Hashem S, Akther S, Jothi JS (2023) Impact of various extraction methods on fatty acid profile, physicochemical properties, and nutritional quality index of Pangus fish oil. Food Sci Nutr 11(8):4688–4699. https://doi.org/10.1002/fsn3.3431
64. Pinela J, Fuente BDL, Rodrigues M, Pires TC, Mandim F, Almeida A, Barros L (2022) Upcycling fish by-products into bioactive fish oil: the suitability of microwave-assisted extraction. Biomolecules 13(1):1. https://doi.org/10.3390/biom13010001
65. Franklin EC, Haq M, Roy VC, Park JS, Chun BS (2020) Supercritical CO2 extraction and quality comparison of lipids from Yellowtail fish (*Seriola quinqueradiata*) waste in different conditions. J Food Process Preserv 44(11):e14892. https://doi.org/10.1111/jfpp.14892
66. Routray W, Dave D, Ramakrishnan VV, Murphy W (2018) Production of high quality fish oil by enzymatic protein hydrolysis from cultured Atlantic salmon by-products: investigation on effect of various extraction parameters using central composite rotatable design. Waste Biomass Valoriz 9(11):2003–2014. https://doi.org/10.1007/s12649-017-9944-1
67. Gnanasekaran L, Priya AK, Thanigaivel S (2023) The conversion of biomass to fuels via cutting-edge technologies: explorations from natural utilization systems. Fuel 331:125668–125681. https://doi.org/10.1016/j.fuel.2022.125668
68. Alalwan HA, Alminshid AH, Aljaafari HA (2019) Promising evolution of biofuel generations. Subject review. Renew Energy Focus 28:127–139. https://doi.org/10.1016/j.ref.2018.12.006
69. Dabirian E, Hajipour A, Mehrizi AA (2023) Nanoparticles application on fuel production from biological resources: a review. Fuel 331:125682–125695. https://doi.org/10.1016/j.fuel.2022.125682
70. Gebremariam SN, Marchetti JM (2017) Biodiesel production technologies. AIMS Energy 5:425–457
71. Pourkarimi S, Hallajisani A, Alizadehdakhel A, Nouralishahi A (2019) Biofuel production through micro-and macroalgae pyrolysis–a review of pyrolysis methods and process parameters. J Anal Appl Pyrolysis 142:104599. https://doi.org/10.1016/j.jaap.2019.04.015
72. Basu P (2018) Biomass gasification, pyrolysis and torrefaction: practical design and theory. Academic
73. Anca-Couce A (2016) Reaction mechanisms and multi-scale modelling of lignocellulosic biomass pyrolysis. Prog Energy Combust Sci 53:41–79. https://doi.org/10.1016/j.pecs.2015.10.002
74. Goldbach A, Meier HF, Wiggers V (2022) Combustion performance of bio-gasoline produced by waste fish oil pyrolysis. Chem Ind Chem Eng Q 28:1–8. https://doi.org/10.2298/CICEQ200810010G
75. Jaiswal KK, Dutta S, Banerjee I, Jaiswal KS, Renuka N, Ratha SK, Jaiswal AK (2024) Valorization of fish processing industry waste for biodiesel production: opportunities, challenges, and technological perspectives. Renew Energy 220:119601. https://doi.org/10.1016/j.renene.2023.119601
76. Karpagam R, Jawaharraj K, Gnanam R (2021) Review on integrated biofuel production from microalgal biomass through the outset of transesterification route: a cascade approach for sustainable bioenergy. Sci Total Environ 766:144236–144256. https://doi.org/10.1016/j.scitotenv.2020.144236
77. Braun JV, Santos SJ, da Costa Espíndola G, Fontoura LAM, Alves AK (2021) Microwave heating and synthesis method influence in SiO 2–ZrO 2 mixed oxides preparation and its use as heterogeneous catalyst for biodiesel obtainment. Reaction Kinetics, Mechanisms and Catalysis 132:921–934. https://doi.org/10.1007/s11144-021-01950-9
78. Marwaha A, Dhir A, Mahla SK, Mohapatra SK (2018) An overview of solid base heterogeneous catalysts for biodiesel production. Catal Rev 60(4):594–628. https://doi.org/10.1080/01614940.2018.1494782
79. Rosson E, Sgarbossa P, Pedrielli F, Mozzon M, Bertani R (2021) Bioliquids from Raw waste animal fats: an alternative renewable energy source. Biomass Convers Biorefin 11:1475–1490. https://doi.org/10.1007/s13399-020-00634-z
80. Maheshwari P, Haider MB, Yusuf M (2022) A review on latest trends in cleaner biodiesel production: role of feedstock, production methods, and catalysts. J Clean Prod 355:131588–131607. https://doi.org/10.1016/j.jclepro.2022.131588
81. Mandari V, Devarai SK (2022) Biodiesel production using homogeneous, heterogeneous, and enzyme catalysts via

transesterification and esterification reactions: a critical review. BioEnergy Res 15(2):935–961. https://doi.org/10.1007/s12155-021-10333-w
82. Kudre TG, Bhaskar N, Sakhare PZ (2017) Optimisation and characterization of biodiesel production from Rohu (*Labeo rohita*) processing waste. Renew Energy 113:1408–1418. https://doi.org/10.1016/j.renene.2017.06.10
83. Gebeyehu KB, Asfaw BT, Abebe T (2020) Review on of a heterogeneous solid base catalyst production from waste animal bone as transesterification of Jatropha oil. J Catal Catal 7(1):13–23
84. Mardhiah HH, Ong HC, Masjuki HH, Lim S, Lee HV (2017) A review on latest developments and future prospects of heterogeneous catalyst in biodiesel production from non-edible oils. Renew Sustain Energy Rev 67:1225–1236. https://doi.org/10.1016/j.rser.2016.09.036
85. Jayakumar M, Karmegam N, Gundupalli MP (2021) Heterogeneous base catalysts: synthesis and application for biodiesel production–a review. Bioresour Technol 331:125054–125066. https://doi.org/10.1016/j.biortech.2021.125054
86. Bucciol F, Colia M, Calcio Gaudino E, Cravotto G (2020) Enabling technologies and sustainable catalysis in biodiesel preparation. Catalysts 10(9):988. https://doi.org/10.3390/catal10090988
87. Sulaiman S, Jamaludin NFA, Kabbashi NA (2019) Development of CaO/PVA catalyst from fish bone for biodiesel production. Bull Chem React Eng Catal 14:153–157. https://doi.org/10.9767/bcrec.14.1.3327.153-157
88. Rajak U, Verma TN (2018) Effect of emission from ethylic biodiesel of edible and non-edible vegetable oil, animal fats, waste oil and alcohol in CI engine. Energy Convers Manage 166:704–718. https://doi.org/10.1016/j.enconman.2018.04.070
89. Fadhil AB, Al-Tikrity ET, Albadree MA (2015) Transesterification of a novel feedstock, *Cyprinus carpio* fish oil: influence of co-solvent and characterization of biodiesel. Fuel 162:215–223. https://doi.org/10.1016/j.fuel.2015.09.001
90. Uemura Y, Tran NT, Naqvi SR (2017) Nano-catalysts for upgrading bio-oil: catalytic decarboxylation and hydrodeoxygenation. In AIP Conference Proceedings 1877: AIP Publishing.https://doi.org/10.1063/1.4999852
91. Moraes PS, Engelmann JI, Igansi AV (2021) Nile tilapia industrialization waste: evaluation of the yield, quality and cost of the biodiesel production process. J Clean Prod 287:125041–125050. https://doi.org/10.1016/j.jclepro.2020.125041
92. Adeoti IA, Hawboldt K (2014) A review of lipid extraction from fish processing by-product for use as a biofuel. Biomass Bioenergy 63:330–340. https://doi.org/10.1016/j.biombioe.2014.02.011
93. Oliver L, Dietrich T, Marañón I, Villarán MC, Barrio RJ (2020) Producing omega-3 polyunsaturated fatty acids: a review of sustainable sources and future trends for the EPA and DHA market. Resources 9(12):148. https://doi.org/10.3390/resources9120148
94. Binhweel F, Hossain MS, Ahmad MI (2023) Recent trends, potentials, and challenges of biodiesel production from discarded animal fats: a comprehensive review. Bioenergy Res 16(2):778–800. https://doi.org/10.1007/s12155-022-10527-w
95. Knothe G (2010) Biodiesel and renewable diesel: a comparison. Prog Energy Combust Sci 36(3):364–373. https://doi.org/10.1016/j.pecs.2009.11.004
96. Atabani AE, Silitonga AS, Ong HC, Mahlia TMI, Masjuki HH, Badruddin IA, Fayaz H (2013) Non-edible vegetable oils: a critical evaluation of oil extraction, fatty acid compositions, biodiesel production, characteristics, engine performance and emissions production. Renew Sustain Energy Rev 18:211–245. https://doi.org/10.1016/j.rser.2012.10.013
97. Ma G, Dai L, Liu D (2018) Lipase-mediated selective methanolysis of fish oil for biodiesel production and polyunsaturated fatty acid enrichment. Energy Fuels 32:7630–7635
98. Eiroa M, Costa JC, Alves MM (2012) Evaluation of the biomethane potential of solid fish waste. Waste Manag 32:1347–1352. https://doi.org/10.1016/j.wasman.2012.03.020
99. Demirbas A (2017) Biofuels from waste biomass: focus on fish processing waste. Renew Sustain Energy Rev 72:1–11. https://doi.org/10.1016/j.rser.2017.01.086
100. Melero JA, Iglesias J, Morales G, Alvarez E (2020) Renewable LPG from waste oils and fats: production technologies and fuel characteristics. Renew Sustain Energy Rev 130:109948. https://doi.org/10.1016/j.rser.2020.109948
101. Hazrat MA, Rasul MG, Mofijur M, Khan MMK, Djavanroodi F, Azad AK, Silitonga AS (2020) A mini review on the cold flow properties of biodiesel and its blends. Front Energy Res 8:598651
102. Malik K, Capareda SC, Kamboj BR (2024) Biofuels production: a review on sustainable alternatives to traditional fuels and energy sources. Fuels 5:157–175. https://doi.org/10.3390/fuels5020010
103. Prado JM, Meireles MAA (2012) Production of valuable compounds by supercritical technology using residues from sugarcane processing.doi/. https://doi.org/10.1002/9780470976692.ch7/pdf
104. Le LT, van Ierland EC, Zhu X, Wesseler J (2013) Energy and greenhouse gas balances of cassava-based ethanol. Biomass Bioenerg 51:125–135. https://doi.org/10.1016/j.biombioe.2013.01.011
105. Cherwoo L, Gupta I, Flora G, Verma R, Kapil M, Arya SK, Ashokkumar V (2023) Biofuels an alternative to traditional fossil fuels: a comprehensive review. Sustain Energy Technol Assess 60:103503. https://doi.org/10.1016/j.seta.2023.103503
106. Shen Sanz B, Albillos Sanchez A, Tangey B, Gilmore K, Yue Z, Liu X, Wallace G (2020) Light cross-linkable marine collagen for coaxial printing of a 3D model of neuromuscular junction formation. Biomedicines 9(1):16–35. https://doi.org/10.3390/biomedicines9010016
107. Ching-Velasquez J, Fernández-Lafuente R, Rodrigues RC, Plata V, Rosales-Quintero A, Torrestiana-Sánchez B, Tacias-Pascacio VG (2020) Production and characterization of biodiesel from oil of fish waste by enzymatic catalysis. Renew Energy 153:1346–1354. https://doi.org/10.1016/j.renene.2020.02.100
108. Ali M, Shakaib M, Zaidi AA, Javed MA, Khan SZ, Aly Hassan A (2025) Performance and environmental sustainability of fish waste biodiesel on diesel engines. Sustainability 17(12):5385. https://doi.org/10.3390/su17125385
109. Kara K, Ouanji F, Lotfi EM, El Mahi M, Kacimi M, Ziyad M (2018) Biodiesel production from waste fish oil with high free fatty acid content from Moroccan fish-processing industries. Egypt J Pet 27(2):249–255. https://doi.org/10.1016/j.ejpe.2017.07.010
110. Yahyaee R, Ghobadian B, Najafi G (2013) Waste fish oil biodiesel as a source of renewable fuel in Iran. Renew Sustain Energy Rev 17:312–319. https://doi.org/10.1016/j.rser.2012.09.025
111. Mohamed MA, Elazab HA, Gadalla MA, Ashour FA (2024) Conversion of Fish Waste Oil into Biofuel: An Experimental Study. https://doi.org/10.33263/LIANBS133.108
112. Milbrandt A, Kinchin C, McCormick R (2013) Feasibility of producing and using biomass-based diesel and jet fuel in the United States (No. NREL/TP-6A20-58015). National Renewable Energy Lab.(NREL), Golden, CO (United States).https://doi.org/10.1016/j.rser.2012.09.025
113. Corvo P (2024) EU-India Stakeholder Group on Advanced Biofuels Lipids and other types of biomass supply chains
114. Dessie W, Luo X, Wang M, Feng L, Liao Y, Wang Z, Qin Z (2020) Current advances on waste biomass transformation into

value-added products. Appl Microbiol Biotechnol 104(11):4757–4770. https://doi.org/10.1007/s00253-020-10567-2. .DOI:
115. Isah S, Ozbay G (2020) Valorization of food loss and wastes: feedstocks for biofuels and valuable chemicals. Front Sustain Food Syst 4:82. https://doi.org/10.3389/fsufs.2020.00082
116. Dhiman S, Mukherjee G (2021) Present scenario and future scope of food waste to biofuel production. J Food Process Eng 44(2):e13594. https://doi.org/10.1111/jfpe.13594

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Esha Razzaq[1] · Syed Makhdoom Hussain[1] · Shafaqat Ali[2] · Ameer Fawad Zahoor[3] · Hamda Azmat[4] · Adan Naeem[1] · Ebru Yilmaz[5] · Eman Naeem[1] · Muhammad Amjad[1]

✉ Syed Makhdoom Hussain
drmakhdoomhussain@gcuf.edu.pk; drmakhdoom90@gmail.com

1 Fish Nutrition Laboratory, Department of Zoology, Government College University, Faisalabad, Faisalabad 38000, Pakistan

2 Department of Environmental Sciences, Government College University, Faisalabad, Faisalabad 38000, Pakistan

3 Department of Chemistry, Government College University Faisalabad, Faisalabad 38000, Pakistan

4 Department of Fisheries and Aquaculture, University of Veterinary and Animal Sciences, Lahore, Pakistan

5 Bozdoğan Vocational School, Aydın Adnan Menderes University, Aydın, Türkiye